B3.2 Transport in plants (HL)

What the Cell · DP Biology · worksheet · HL statements B3.2.17–18

Name _____ **Date** _____ **Total** _____ / 20

About 20 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

5 marks

1 **Define** translocation. B3.2.18 **[1]**

2 **State** the name for drops of xylem sap pushed out at the edges of leaves by root pressure. B3.2.17 **[1]**

3 **State** two features of companion cells. B3.2.18 **[2]**

4 **Name** one part of a plant that acts as a sink. B3.2.18 **[1]**

Section B: Understanding

9 marks

5 **Explain** how root pressure is generated. B3.2.17 **[3]**

6 **Explain** how sieve tube elements are adapted for the flow of sap. B3.2.18 **[3]**

7 **Suggest** why root pressure matters more to a deciduous tree in early spring than in summer. B3.2.17 **[3]**

Section C: Working with evidence

6 marks

All data in this section are invented for practice. The patterns are realistic.

8 B3.2.17

The shoots of two groups of tomato plants were cut off just above the soil, and the sap that came out of each stump was collected. After 1 hour, a respiratory inhibitor was added to the water around the roots of one group. The table shows the mean total volume of sap collected.

Time / h	**0**	**1**	**2**	**3**	**4**
No inhibitor / cm^3	0	0.42	0.85	1.26	1.70
Inhibitor added at 1 h / cm^3	0	0.40	0.55	0.58	0.59

a **Calculate** the mean rate of sap flow from the plants with no inhibitor over the 4 hours. **[1]**

b **Describe** the effect of the inhibitor. **[2]**

c **Explain** the effect of the inhibitor. **[3]**

What the Cell. Original questions written for this course.